

9.1 Basic configuration

GPIOs are configured using the following registers:

1. Power: always enabled.
2. Pins: See [Section 8.3](#) for GPIO pins and their modes.
3. Wake-up: GPIO ports 0 and 2 can be used for wake-up if needed, see ([Section 4.8.8](#)).
4. Interrupts: Enable GPIO interrupts in IO0/2IntEnR ([Table 115](#)) or IO0/2IntEnF ([Table 117](#)). Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register.

9.2 Features

9.2.1 Digital I/O ports

- Accelerated GPIO functions:
 - GPIO registers are located on a peripheral AHB bus for fast I/O timing.
 - Mask registers allow treating sets of port bits as a group, leaving other bits unchanged.
 - All GPIO registers are byte, half-word, and word addressable.
 - Entire port value can be written in one instruction.
 - GPIO registers are accessible by the GPDMA.
- Bit-level set and clear registers allow a single instruction set or clear of any number of bits in one port.
- All GPIO registers support Cortex-M3 bit-banding.
- GPIO registers are accessible by the GPDMA controller to allow DMA of data to or from GPIOs, synchronized to any DMA request.
- Direction control of individual port bits.
- All I/Os default to input with pullup after reset.

9.2.2 Interrupt generating digital ports

- Port 0 and Port 2 can provide a single interrupt for any combination of port pins.
- Each port pin can be programmed to generate an interrupt on a rising edge, a falling edge, or both.
- Edge detection is asynchronous, so it may operate when clocks are not present, such as during Power-down mode. With this feature, level triggered interrupts are not needed.
- Each enabled interrupt contributes to a wake-up signal that can be used to bring the part out of Power-down mode.

- Registers provide a software view of pending rising edge interrupts, pending falling edge interrupts, and overall pending GPIO interrupts.
- GPIO0 and GPIO2 interrupts share the same position in the NVIC with External Interrupt 3.

9.3 Applications

- General purpose I/O
- Driving LEDs or other indicators
- Controlling off-chip devices
- Sensing digital inputs, detecting edges
- Bringing the part out of Power-down mode

9.4 Pin description

Table 101. GPIO pin description

Pin Name	Type	Description
P0[30:0] ^[1] ; P1[31:0] ^[2] ; P2[13:0]; P3[26:25]; P4[29:28]	Input/ Output	General purpose input/output. These are typically shared with other peripherals functions and will therefore not all be available in an application. Packaging options may affect the number of GPIOs available in a particular device. Some pins may be limited by requirements of the alternate functions of the pin. For example, the pins containing the I ² C0 functions are open-drain for any function selected on that pin. Details may be found in Section 7.1.1 .

[1] P0[14:12] are not available.

[2] P1[2], P1[3], P1[7:5], P1[13:11] are not available.

9.5 Register description

Due to compatibility requirements with the LPC2300 series ARM7-based products, the LPC176x/5x implements portions of five 32-bit General Purpose I/O ports. Details on a specific GPIO port usage can be found in [Section 8.3](#).

The registers in [Table 102](#) represent the enhanced GPIO features available on all of the GPIO ports. These registers are located on an AHB bus for fast read and write timing. They can all be accessed in byte, half-word, and word sizes. A mask register allows access to a group of bits in a single GPIO port independently from other bits in the same port.

Table 102. GPIO register map (local bus accessible registers - enhanced GPIO features)

Generic Name	Description	Access	Reset value ^[1]	PORTn Register Name & Address
FIODIR	Fast GPIO Port Direction control register. This register individually controls the direction of each port pin.	R/W	0	FIO0DIR - 0x2009 C000 FIO1DIR - 0x2009 C020 FIO2DIR - 0x2009 C040 FIO3DIR - 0x2009 C060 FIO4DIR - 0x2009 C080
FIOMASK	Fast Mask register for port. Writes, sets, clears, and reads to port (done via writes to FIOPIN, FIOSET, and FIOCLR, and reads of FIOPIN) alter or return only the bits enabled by zeros in this register.	R/W	0	FIO0MASK - 0x2009 C010 FIO1MASK - 0x2009 C030 FIO2MASK - 0x2009 C050 FIO3MASK - 0x2009 C070 FIO4MASK - 0x2009 C090
FIOPIN	Fast Port Pin value register using FIOMASK. The current state of digital port pins can be read from this register, regardless of pin direction or alternate function selection (as long as pins are not configured as an input to ADC). The value read is masked by ANDing with inverted FIOMASK. Writing to this register places corresponding values in all bits enabled by zeros in FIOMASK. Important: if an FIOPIN register is read, its bit(s) masked with 1 in the FIOMASK register will be read as 0 regardless of the physical pin state.	R/W	0	FIO0PIN - 0x2009 C014 FIO1PIN - 0x2009 C034 FIO2PIN - 0x2009 C054 FIO3PIN - 0x2009 C074 FIO4PIN - 0x2009 C094
FIOSET	Fast Port Output Set register using FIOMASK. This register controls the state of output pins. Writing 1s produces highs at the corresponding port pins. Writing 0s has no effect. Reading this register returns the current contents of the port output register. Only bits enabled by 0 in FIOMASK can be altered.	R/W	0	FIO0SET - 0x2009 C018 FIO1SET - 0x2009 C038 FIO2SET - 0x2009 C058 FIO3SET - 0x2009 C078 FIO4SET - 0x2009 C098
FIOCLR	Fast Port Output Clear register using FIOMASK. This register controls the state of output pins. Writing 1s produces lows at the corresponding port pins. Writing 0s has no effect. Only bits enabled by 0 in FIOMASK can be altered.	WO	0	FIO0CLR - 0x2009 C01C FIO1CLR - 0x2009 C03C FIO2CLR - 0x2009 C05C FIO3CLR - 0x2009 C07C FIO4CLR - 0x2009 C09C

[1] Reset value reflects the data stored in used bits only. It does not include reserved bits content.

Table 103. GPIO interrupt register map

Generic Name	Description	Access	Reset value ^[1]	PORTn Register Name & Address
IntEnR	GPIO Interrupt Enable for Rising edge.	R/W	0	IO0IntEnR - 0x4002 8090 IO2IntEnR - 0x4002 80B0
IntEnF	GPIO Interrupt Enable for Falling edge.	R/W	0	IO0IntEnR - 0x4002 8094 IO2IntEnR - 0x4002 80B4
IntStatR	GPIO Interrupt Status for Rising edge.	RO	0	IO0IntStatR - 0x4002 8084 IO2IntStatR - 0x4002 80A4
IntStatF	GPIO Interrupt Status for Falling edge.	RO	0	IO0IntStatF - 0x4002 8088 IO2IntStatF - 0x4002 80A8
IntClr	GPIO Interrupt Clear.	WO	0	IO0IntClr - 0x4002 808C IO2IntClr - 0x4002 80AC
IntStatus	GPIO overall Interrupt Status.	RO	0	IOIntStatus - 0x4002 8080

[1] Reset value reflects the data stored in used bits only. It does not include reserved bits content.

9.5.1 GPIO port Direction register FIOxDIR (FIO0DIR to FIO4DIR- 0x2009 C000 to 0x2009 C080)

This word accessible register is used to control the direction of the pins when they are configured as GPIO port pins. Direction bit for any pin must be set according to the pin functionality.

Note that GPIO pins P0.29 and P0.30 are shared with the USB_D+ and USB_D- pins and must have the same direction. If either FIO0DIR bit 29 or 30 are configured as zero, both P0.29 and P0.30 will be inputs. If both FIO0DIR bits 29 and 30 are ones, both P0.29 and P0.30 will be outputs.

Table 104. Fast GPIO port Direction register FIO0DIR to FIO4DIR - addresses 0x2009 C000 to 0x2009 C080) bit description

Bit	Symbol	Value	Description	Reset value
31:0	FIO0DIR FIO1DIR FIO2DIR FIO3DIR FIO4DIR		Fast GPIO Direction PORTx control bits. Bit 0 in FIOxDIR controls pin Px.0, bit 31 in FIOxDIR controls pin Px.31.	0x0
		0	Controlled pin is input.	
		1	Controlled pin is output.	

Aside from the 32-bit long and word only accessible FIODIR register, every fast GPIO port can also be controlled via several byte and half-word accessible registers listed in [Table 105](#), too. Next to providing the same functions as the FIODIR register, these additional registers allow easier and faster access to the physical port pins.

Table 105. Fast GPIO port Direction control byte and half-word accessible register description

Generic Register name	Description	Register length (bits) & access	Reset value	PORTn Register Address & Name
FIOxDIR0	Fast GPIO Port x Direction control register 0. Bit 0 in FIOxDIR0 register corresponds to pin Px.0 ... bit 7 to pin Px.7.	8 (byte) R/W	0x00	FIO0DIR0 - 0x2009 C000 FIO1DIR0 - 0x2009 C020 FIO2DIR0 - 0x2009 C040 FIO3DIR0 - 0x2009 C060 FIO4DIR0 - 0x2009 C080
FIOxDIR1	Fast GPIO Port x Direction control register 1. Bit 0 in FIOxDIR1 register corresponds to pin Px.8 ... bit 7 to pin Px.15.	8 (byte) R/W	0x00	FIO0DIR1 - 0x2009 C001 FIO1DIR1 - 0x2009 C021 FIO2DIR1 - 0x2009 C041 FIO3DIR1 - 0x2009 C061 FIO4DIR1 - 0x2009 C081
FIOxDIR2	Fast GPIO Port x Direction control register 2. Bit 0 in FIOxDIR2 register corresponds to pin Px.16 ... bit 7 to pin Px.23.	8 (byte) R/W	0x00	FIO0DIR2 - 0x2009 C002 FIO1DIR2 - 0x2009 C022 FIO2DIR2 - 0x2009 C042 FIO3DIR2 - 0x2009 C062 FIO4DIR2 - 0x2009 C082
FIOxDIR3	Fast GPIO Port x Direction control register 3. Bit 0 in FIOxDIR3 register corresponds to pin Px.24 ... bit 7 to pin Px.31.	8 (byte) R/W	0x00	FIO0DIR3 - 0x2009 C003 FIO1DIR3 - 0x2009 C023 FIO2DIR3 - 0x2009 C043 FIO3DIR3 - 0x2009 C063 FIO4DIR3 - 0x2009 C083
FIOxDIRL	Fast GPIO Port x Direction control Lower half-word register. Bit 0 in FIOxDIRL register corresponds to pin Px.0 ... bit 15 to pin Px.15.	16 (half-word) R/W	0x0000	FIO0DIRL - 0x2009 C000 FIO1DIRL - 0x2009 C020 FIO2DIRL - 0x2009 C040 FIO3DIRL - 0x2009 C060 FIO4DIRL - 0x2009 C080
FIOxDIRU	Fast GPIO Port x Direction control Upper half-word register. Bit 0 in FIOxDIRU register corresponds to Px.16 ... bit 15 to Px.31.	16 (half-word) R/W	0x0000	FIO0DIRU - 0x2009 C002 FIO1DIRU - 0x2009 C022 FIO2DIRU - 0x2009 C042 FIO3DIRU - 0x2009 C062 FIO4DIRU - 0x2009 C082

9.5.2 GPIO port output Set register FIOxSET (FIO0SET to FIO4SET - 0x2009 C018 to 0x2009 C098)

This register is used to produce a HIGH level output at the port pins configured as GPIO in an OUTPUT mode. Writing 1 produces a HIGH level at the corresponding port pins. Writing 0 has no effect. If any pin is configured as an input or a secondary function, writing 1 to the corresponding bit in the FIOxSET has no effect.

Reading the FIOxSET register returns the value of this register, as determined by previous writes to FIOxSET and FIOxCLR (or FIOxPIN as noted above). This value does not reflect the effect of any outside world influence on the I/O pins.

Access to a port pin via the FIOxSET register is conditioned by the corresponding bit of the FIOxMASK register (see [Section 9.5.5](#)).

Table 106. Fast GPIO port output Set register (FIO0SET to FIO4SET - addresses 0x2009 C018 to 0x2009 C098) bit description

Bit	Symbol	Value	Description	Reset value
31:0	FIO0SET FIO1SET FIO2SET FIO3SET FIO4SET	0 1	Fast GPIO output value Set bits. Bit 0 in FIOxSET controls pin Px.0, bit 31 in FIOxSET controls pin Px.31. Controlled pin output is unchanged. Controlled pin output is set to HIGH.	0x0

Aside from the 32-bit long and word only accessible FIOxSET register, every fast GPIO port can also be controlled via several byte and half-word accessible registers listed in [Table 107](#), too. Next to providing the same functions as the FIOxSET register, these additional registers allow easier and faster access to the physical port pins.

Table 107. Fast GPIO port output Set byte and half-word accessible register description

Generic Register name	Description	Register length (bits) & access	Reset value	PORTn Register Address & Name
FIOxSET0	Fast GPIO Port x output Set register 0. Bit 0 in FIOxSET0 register corresponds to pin Px.0 ... bit 7 to pin Px.7.	8 (byte) R/W	0x00	FIO0SET0 - 0x2009 C018 FIO1SET0 - 0x2009 C038 FIO2SET0 - 0x2009 C058 FIO3SET0 - 0x2009 C078 FIO4SET0 - 0x2009 C098
FIOxSET1	Fast GPIO Port x output Set register 1. Bit 0 in FIOxSET1 register corresponds to pin Px.8 ... bit 7 to pin Px.15.	8 (byte) R/W	0x00	FIO0SET1 - 0x2009 C019 FIO1SET1 - 0x2009 C039 FIO2SET1 - 0x2009 C059 FIO3SET1 - 0x2009 C079 FIO4SET1 - 0x2009 C099
FIOxSET2	Fast GPIO Port x output Set register 2. Bit 0 in FIOxSET2 register corresponds to pin Px.16 ... bit 7 to pin Px.23.	8 (byte) R/W	0x00	FIO0SET2 - 0x2009 C01A FIO1SET2 - 0x2009 C03A FIO2SET2 - 0x2009 C05A FIO3SET2 - 0x2009 C07A FIO4SET2 - 0x2009 C09A
FIOxSET3	Fast GPIO Port x output Set register 3. Bit 0 in FIOxSET3 register corresponds to pin Px.24 ... bit 7 to pin Px.31.	8 (byte) R/W	0x00	FIO0SET3 - 0x2009 C01B FIO1SET3 - 0x2009 C03B FIO2SET3 - 0x2009 C05B FIO3SET3 - 0x2009 C07B FIO4SET3 - 0x2009 C09B
FIOxSETL	Fast GPIO Port x output Set Lower half-word register. Bit 0 in FIOxSETL register corresponds to pin Px.0 ... bit 15 to pin Px.15.	16 (half-word) R/W	0x0000	FIO0SETL - 0x2009 C018 FIO1SETL - 0x2009 C038 FIO2SETL - 0x2009 C058 FIO3SETL - 0x2009 C078 FIO4SETL - 0x2009 C098
FIOxSETU	Fast GPIO Port x output Set Upper half-word register. Bit 0 in FIOxSETU register corresponds to Px.16 ... bit 15 to Px.31.	16 (half-word) R/W	0x0000	FIO0SETU - 0x2009 C01A FIO1SETU - 0x2009 C03A FIO2SETU - 0x2009 C05A FIO3SETU - 0x2009 C07A FIO4SETU - 0x2009 C09A

9.5.3 GPIO port output Clear register FIOxCLR (FIO0CLR to FIO4CLR-0x2009 C01C to 0x2009 C09C)

This register is used to produce a LOW level output at port pins configured as GPIO in an OUTPUT mode. Writing 1 produces a LOW level at the corresponding port pin and clears the corresponding bit in the FIOxSET register. Writing 0 has no effect. If any pin is configured as an input or a secondary function, writing to FIOxCLR has no effect.

Access to a port pin via the FIOxCLR register is conditioned by the corresponding bit of the FIOxMASK register (see [Section 9.5.5](#)).

Table 108. Fast GPIO port output Clear register (FIO0CLR to FIO4CLR- addresses 0x2009 C01C to 0x2009 C09C) bit description

Bit	Symbol	Value	Description	Reset value
31:0	FIO0CLR FIO1CLR FIO2CLR FIO3CLR FIO4CLR		Fast GPIO output value Clear bits. Bit 0 in FIOxCLR controls pin Px.0, bit 31 controls pin Px.31.	0x0
		0	Controlled pin output is unchanged.	
		1	Controlled pin output is set to LOW.	

Aside from the 32-bit long and word only accessible FIOxCLR register, every fast GPIO port can also be controlled via several byte and half-word accessible registers listed in [Table 109](#), too. Next to providing the same functions as the FIOxCLR register, these additional registers allow easier and faster access to the physical port pins.

Table 109. Fast GPIO port output Clear byte and half-word accessible register description

Generic Register name	Description	Register length (bits) & access	Reset value	PORTn Register Address & Name
FIOxCLR0	Fast GPIO Port x output Clear register 0. Bit 0 in FIOxCLR0 register corresponds to pin Px.0 ... bit 7 to pin Px.7.	8 (byte) WO	0x00	FIO0CLR0 - 0x2009 C01C FIO1CLR0 - 0x2009 C03C FIO2CLR0 - 0x2009 C05C FIO3CLR0 - 0x2009 C07C FIO4CLR0 - 0x2009 C09C
FIOxCLR1	Fast GPIO Port x output Clear register 1. Bit 0 in FIOxCLR1 register corresponds to pin Px.8 ... bit 7 to pin Px.15.	8 (byte) WO	0x00	FIO0CLR1 - 0x2009 C01D FIO1CLR1 - 0x2009 C03D FIO2CLR1 - 0x2009 C05D FIO3CLR1 - 0x2009 C07D FIO4CLR1 - 0x2009 C09D
FIOxCLR2	Fast GPIO Port x output Clear register 2. Bit 0 in FIOxCLR2 register corresponds to pin Px.16 ... bit 7 to pin Px.23.	8 (byte) WO	0x00	FIO0CLR2 - 0x2009 C01E FIO1CLR2 - 0x2009 C03E FIO2CLR2 - 0x2009 C05E FIO3CLR2 - 0x2009 C07E FIO4CLR2 - 0x2009 C09E

Table 109. Fast GPIO port output Clear byte and half-word accessible register description ...continued

Generic Register name	Description	Register length (bits) & access	Reset value	PORTn Register Address & Name
FIOxCLR3	Fast GPIO Port x output Clear register 3. Bit 0 in FIOxCLR3 register corresponds to pin Px.24 ... bit 7 to pin Px.31.	8 (byte) WO	0x00	FIO0CLR3 - 0x2009 C01F FIO1CLR3 - 0x2009 C03F FIO2CLR3 - 0x2009 C05F FIO3CLR3 - 0x2009 C07F FIO4CLR3 - 0x2009 C09F
FIOxCLRL	Fast GPIO Port x output Clear Lower half-word register. Bit 0 in FIOxCLRL register corresponds to pin Px.0 ... bit 15 to pin Px.15.	16 (half-word) WO	0x0000	FIO0CLRL - 0x2009 C01C FIO1CLRL - 0x2009 C03C FIO2CLRL - 0x2009 C05C FIO3CLRL - 0x2009 C07C FIO4CLRL - 0x2009 C09C
FIOxCLRU	Fast GPIO Port x output Clear Upper half-word register. Bit 0 in FIOxCLRU register corresponds to pin Px.16 ... bit 15 to Px.31.	16 (half-word) WO	0x0000	FIO0CLRU - 0x2009 C01E FIO1CLRU - 0x2009 C03E FIO2CLRU - 0x2009 C05E FIO3CLRU - 0x2009 C07E FIO4CLRU - 0x2009 C09E

9.5.4 GPIO port Pin value register FIOxPIN (FIO0PIN to FIO4PIN- 0x2009 C014 to 0x2009 C094)

This register provides the value of port pins that are configured to perform only digital functions. The register will give the logic value of the pin regardless of whether the pin is configured for input or output, or as GPIO or an alternate digital function. As an example, a particular port pin may have GPIO input, GPIO output, UART receive, and PWM output as selectable functions. Any configuration of that pin will allow its current logic state to be read from the corresponding FIOxPIN register.

If a pin has an analog function as one of its options, the pin state cannot be read if the analog configuration is selected. Selecting the pin as an A/D input disconnects the digital features of the pin. In that case, the pin value read in the FIOxPIN register is not valid.

Writing to the FIOxPIN register stores the value in the port output register, bypassing the need to use both the FIOxSET and FIOxCLR registers to obtain the entire written value. This feature should be used carefully in an application since it affects the entire port.

Access to a port pin via the FIOxPIN register is conditioned by the corresponding bit of the FIOxMASK register (see [Section 9.5.5](#)).

Only pins masked with zeros in the Mask register (see [Section 9.5.5](#)) will be correlated to the current content of the Fast GPIO port pin value register.

Table 110. Fast GPIO port Pin value register (FIO0PIN to FIO4PIN- addresses 0x2009 C014 to 0x2009 C094) bit description

Bit	Symbol	Value	Description	Reset value
31:0	FIO0VAL FIO1VAL FIO2VAL FIO3VAL FIO4VAL		Fast GPIO output value bits. Bit 0 corresponds to pin Px.0, bit 31 corresponds to pin Px.31. Only bits also set to 0 in the FIOxMASK register are affected by a write or show the pin's actual logic state.	0x0
		0	Reading a 0 indicates that the port pin's current state is LOW. Writing a 0 sets the output register value to LOW.	
		1	Reading a 1 indicates that the port pin's current state is HIGH. Writing a 1 sets the output register value to HIGH.	

Aside from the 32-bit long and word only accessible FIOxPIN register, every fast GPIO port can also be controlled via several byte and half-word accessible register listed in [Table 111](#), too. Next to providing the same functions as the FIOxPIN register, these additional registers allow easier and faster access to the physical port pins.

Table 111. Fast GPIO port Pin value byte and half-word accessible register description

Generic Register name	Description	Register length (bits) & access	Reset value	PORTn Register Address & Name
FIOxPIN0	Fast GPIO Port x Pin value register 0. Bit 0 in FIOxPIN0 register corresponds to pin Px.0 ... bit 7 to pin Px.7.	8 (byte) R/W	0x00	FIO0PIN0 - 0x2009 C014 FIO1PIN0 - 0x2009 C034 FIO2PIN0 - 0x2009 C054 FIO3PIN0 - 0x2009 C074 FIO4PIN0 - 0x2009 C094
FIOxPIN1	Fast GPIO Port x Pin value register 1. Bit 0 in FIOxPIN1 register corresponds to pin Px.8 ... bit 7 to pin Px.15.	8 (byte) R/W	0x00	FIO0PIN1 - 0x2009 C015 FIO1PIN1 - 0x2009 C035 FIO2PIN1 - 0x2009 C055 FIO3PIN1 - 0x2009 C075 FIO4PIN1 - 0x2009 C095
FIOxPIN2	Fast GPIO Port x Pin value register 2. Bit 0 in FIOxPIN2 register corresponds to pin Px.16 ... bit 7 to pin Px.23.	8 (byte) R/W	0x00	FIO0PIN2 - 0x2009 C016 FIO1PIN2 - 0x2009 C036 FIO2PIN2 - 0x2009 C056 FIO3PIN2 - 0x2009 C076 FIO4PIN2 - 0x2009 C096

Table 111. Fast GPIO port Pin value byte and half-word accessible register description ...continued

Generic Register name	Description	Register length (bits) & access	Reset value	PORTn Register Address & Name
FIOxPIN3	Fast GPIO Port x Pin value register 3. Bit 0 in FIOxPIN3 register corresponds to pin Px.24 ... bit 7 to pin Px.31.	8 (byte) R/W	0x00	FIO0PIN3 - 0x2009 C017 FIO1PIN3 - 0x2009 C037 FIO2PIN3 - 0x2009 C057 FIO3PIN3 - 0x2009 C077 FIO4PIN3 - 0x2009 C097
FIOxPINL	Fast GPIO Port x Pin value Lower half-word register. Bit 0 in FIOxPINL register corresponds to pin Px.0 ... bit 15 to pin Px.15.	16 (half-word) R/W	0x0000	FIO0PINL - 0x2009 C014 FIO1PINL - 0x2009 C034 FIO2PINL - 0x2009 C054 FIO3PINL - 0x2009 C074 FIO4PINL - 0x2009 C094
FIOxPINU	Fast GPIO Port x Pin value Upper half-word register. Bit 0 in FIOxPINU register corresponds to pin Px.16 ... bit 15 to Px.31.	16 (half-word) R/W	0x0000	FIO0PINU - 0x2009 C016 FIO1PINU - 0x2009 C036 FIO2PINU - 0x2009 C056 FIO3PINU - 0x2009 C076 FIO4PINU - 0x2009 C096

9.5.5 Fast GPIO port Mask register FIOxMASK (FIO0MASK to FIO4MASK - 0x2009 C010 to 0x2009 C090)

This register is used to select port pins that will and will not be affected by write accesses to the FIOxPIN, FIOxSET or FIOxCLR register. Mask register also filters out port's content when the FIOxPIN register is read.

A zero in this register's bit enables an access to the corresponding physical pin via a read or write access. If a bit in this register is one, corresponding pin will not be changed with write access and if read, will not be reflected in the updated FIOxPIN register. For software examples, see [Section 9.6](#).

Table 112. Fast GPIO port Mask register (FIO0MASK to FIO4MASK - addresses 0x2009 C010 to 0x2009 C090) bit description

Bit	Symbol	Value	Description	Reset value
31:0	FIO0MASK FIO1MASK FIO2MASK FIO3MASK FIO4MASK	0	Fast GPIO physical pin access control. Controlled pin is affected by writes to the port's FIOxSET, FIOxCLR, and FIOxPIN register(s). Current state of the pin can be read from the FIOxPIN register.	0x0
		1	Controlled pin is not affected by writes into the port's FIOxSET, FIOxCLR and FIOxPIN register(s). When the FIOxPIN register is read, this bit will not be updated with the state of the physical pin.	

Aside from the 32-bit long and word only accessible FIOxMASK register, every fast GPIO port can also be controlled via several byte and half-word accessible registers listed in [Table 113](#), too. Next to providing the same functions as the FIOxMASK register, these additional registers allow easier and faster access to the physical port pins.

Table 113. Fast GPIO port Mask byte and half-word accessible register description

Generic Register name	Description	Register length (bits) & access	Reset value	PORTn Register Address & Name
FIOxMASK0	Fast GPIO Port x Mask register 0. Bit 0 in FIOxMASK0 register corresponds to pin Px.0 ... bit 7 to pin Px.7.	8 (byte) R/W	0x0	FIO0MASK0 - 0x2009 C010 FIO1MASK0 - 0x2009 C030 FIO2MASK0 - 0x2009 C050 FIO3MASK0 - 0x2009 C070 FIO4MASK0 - 0x2009 C090
FIOxMASK1	Fast GPIO Port x Mask register 1. Bit 0 in FIOxMASK1 register corresponds to pin Px.8 ... bit 7 to pin Px.15.	8 (byte) R/W	0x0	FIO0MASK1 - 0x2009 C011 FIO1MASK1 - 0x2009 C031 FIO2MASK1 - 0x2009 C051 FIO3MASK1 - 0x2009 C071 FIO4MASK1 - 0x2009 C091
FIOxMASK2	Fast GPIO Port x Mask register 2. Bit 0 in FIOxMASK2 register corresponds to pin Px.16 ... bit 7 to pin Px.23.	8 (byte) R/W	0x0	FIO0MASK2 - 0x2009 C012 FIO1MASK2 - 0x2009 C032 FIO2MASK2 - 0x2009 C052 FIO3MASK2 - 0x2009 C072 FIO4MASK2 - 0x2009 C092
FIOxMASK3	Fast GPIO Port x Mask register 3. Bit 0 in FIOxMASK3 register corresponds to pin Px.24 ... bit 7 to pin Px.31.	8 (byte) R/W	0x0	FIO0MASK3 - 0x2009 C013 FIO1MASK3 - 0x2009 C033 FIO2MASK3 - 0x2009 C053 FIO3MASK3 - 0x2009 C073 FIO4MASK3 - 0x2009 C093
FIOxMASKL	Fast GPIO Port x Mask Lower half-word register. Bit 0 in FIOxMASKL register corresponds to pin Px.0 ... bit 15 to pin Px.15.	16 (half-word) R/W	0x0	FIO0MASKL - 0x2009 C010 FIO1MASKL - 0x2009 C030 FIO2MASKL - 0x2009 C050 FIO3MASKL - 0x2009 C070 FIO4MASKL - 0x2009 C090
FIOxMASKU	Fast GPIO Port x Mask Upper half-word register. Bit 0 in FIOxMASKU register corresponds to pin Px.16 ... bit 15 to Px.31.	16 (half-word) R/W	0x0	FIO0MASKU - 0x2009 C012 FIO1MASKU - 0x2009 C032 FIO2MASKU - 0x2009 C052 FIO3MASKU - 0x2009 C072 FIO4MASKU - 0x2009 C092

9.5.6 GPIO interrupt registers

The following registers configure the pins of Port 0 and Port 2 to generate interrupts.

9.5.6.1 GPIO overall Interrupt Status register (IOIntStatus - 0x4002 8080)

This read-only register indicates the presence of interrupt pending on all of the GPIO ports that support GPIO interrupts. Only status one bit per port is required.

Table 114. GPIO overall Interrupt Status register (IOIntStatus - address 0x4002 8080) bit description

Bit	Symbol	Value	Description	Reset value
0	P0Int		Port 0 GPIO interrupt pending.	0
		0	There are no pending interrupts on Port 0.	
		1	There is at least one pending interrupt on Port 0.	
1	-	-	Reserved. The value read from a reserved bit is not defined.	NA
2	P2Int		Port 2 GPIO interrupt pending.	0
		0	There are no pending interrupts on Port 2.	
		1	There is at least one pending interrupt on Port 2.	
31:2	-	-	Reserved. The value read from a reserved bit is not defined.	NA

9.5.6.2 GPIO Interrupt Enable for port 0 Rising Edge (IO0IntEnR - 0x4002 8090)

Each bit in these read-write registers enables the rising edge interrupt for the corresponding port 0 pin.

Table 115. GPIO Interrupt Enable for port 0 Rising Edge (IO0IntEnR - 0x4002 8090) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0ER		Enable rising edge interrupt for P0.0.	0
		0	Rising edge interrupt is disabled on P0.0.	
		1	Rising edge interrupt is enabled on P0.0.	
1	P0.1ER		Enable rising edge interrupt for P0.1.	0
2	P0.2ER		Enable rising edge interrupt for P0.2.	0
3	P0.3ER		Enable rising edge interrupt for P0.3.	0
4	P0.4ER ^[1]		Enable rising edge interrupt for P0.4.	0
5	P0.5ER ^[1]		Enable rising edge interrupt for P0.5.	0
6	P0.6ER		Enable rising edge interrupt for P0.6.	0
7	P0.7ER		Enable rising edge interrupt for P0.7.	0
8	P0.8ER		Enable rising edge interrupt for P0.8.	0
9	P0.9ER		Enable rising edge interrupt for P0.9.	0
10	P0.10ER		Enable rising edge interrupt for P0.10.	0
11	P0.11ER		Enable rising edge interrupt for P0.11.	0
14:12	-		Reserved	NA
15	P0.15ER		Enable rising edge interrupt for P0.15.	0
16	P0.16ER		Enable rising edge interrupt for P0.16.	0

Table 115. GPIO Interrupt Enable for port 0 Rising Edge (IO0IntEnR - 0x4002 8090) bit description ...continued

Bit	Symbol	Value	Description	Reset value
17	P0.17ER		Enable rising edge interrupt for P0.17.	0
18	P0.18ER		Enable rising edge interrupt for P0.18.	0
19	P0.19ER ^[1]		Enable rising edge interrupt for P0.19.	0
20	P0.20ER ^[1]		Enable rising edge interrupt for P0.20.	0
21	P0.21ER ^[1]		Enable rising edge interrupt for P0.21.	0
22	P0.22ER		Enable rising edge interrupt for P0.22.	0
23	P0.23ER ^[1]		Enable rising edge interrupt for P0.23.	0
24	P0.24ER ^[1]		Enable rising edge interrupt for P0.24.	0
25	P0.25ER		Enable rising edge interrupt for P0.25.	0
26	P0.26ER		Enable rising edge interrupt for P0.26.	0
27	P0.27ER ^[1]		Enable rising edge interrupt for P0.27.	0
28	P0.28ER ^[1]		Enable rising edge interrupt for P0.28.	0
29	P0.29ER		Enable rising edge interrupt for P0.29.	0
30	P0.30ER		Enable rising edge interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.3 GPIO Interrupt Enable for port 2 Rising Edge (IO2IntEnR - 0x4002 80B0)

Each bit in these read-write registers enables the rising edge interrupt for the corresponding port 2 pin.

Table 116. GPIO Interrupt Enable for port 2 Rising Edge (IO2IntEnR - 0x4002 80B0) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0ER		Enable rising edge interrupt for P2.0.	0
		0	Rising edge interrupt is disabled on P2.0.	
		1	Rising edge interrupt is enabled on P2.0.	
1	P2.1ER		Enable rising edge interrupt for P2.1.	0
2	P2.2ER		Enable rising edge interrupt for P2.2.	0
3	P2.3ER		Enable rising edge interrupt for P2.3.	0
4	P2.4ER		Enable rising edge interrupt for P2.4.	0
5	P2.5ER		Enable rising edge interrupt for P2.5.	0
6	P2.6ER		Enable rising edge interrupt for P2.6.	0
7	P2.7ER		Enable rising edge interrupt for P2.7.	0
8	P2.8ER		Enable rising edge interrupt for P2.8.	0
9	P2.9ER		Enable rising edge interrupt for P2.9.	0
10	P2.10ER		Enable rising edge interrupt for P2.10.	0
11	P2.11ER ^[1]		Enable rising edge interrupt for P2.11.	0

Table 116. GPIO Interrupt Enable for port 2 Rising Edge (IO2IntEnR - 0x4002 80B0) bit description ...continued

Bit	Symbol	Value	Description	Reset value
12	P2.12ER ^[1]		Enable rising edge interrupt for P2.12.	0
13	P2.13ER ^[1]		Enable rising edge interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.4 GPIO Interrupt Enable for port 0 Falling Edge (IO0IntEnF - 0x4002 8094)

Each bit in these read-write registers enables the falling edge interrupt for the corresponding GPIO port 0 pin.

Table 117. GPIO Interrupt Enable for port 0 Falling Edge (IO0IntEnF - address 0x4002 8094) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0EF		Enable falling edge interrupt for P0.0	0
		0	Falling edge interrupt is disabled on P0.0.	
		1	Falling edge interrupt is enabled on P0.0.	
1	P0.1EF		Enable falling edge interrupt for P0.1.	0
2	P0.2EF		Enable falling edge interrupt for P0.2.	0
3	P0.3EF		Enable falling edge interrupt for P0.3.	0
4	P0.4EF ^[1]		Enable falling edge interrupt for P0.4.	0
5	P0.5EF ^[1]		Enable falling edge interrupt for P0.5.	0
6	P0.6EF		Enable falling edge interrupt for P0.6.	0
7	P0.7EF		Enable falling edge interrupt for P0.7.	0
8	P0.8EF		Enable falling edge interrupt for P0.8.	0
9	P0.9EF		Enable falling edge interrupt for P0.9.	0
10	P0.10EF		Enable falling edge interrupt for P0.10.	0
11	P0.11EF		Enable falling edge interrupt for P0.11.	0
14:12	-		Reserved.	NA
15	P0.15EF		Enable falling edge interrupt for P0.15.	0
16	P0.16EF		Enable falling edge interrupt for P0.16.	0
17	P0.17EF		Enable falling edge interrupt for P0.17.	0
18	P0.18EF		Enable falling edge interrupt for P0.18.	0
19	P0.19EF ^[1]		Enable falling edge interrupt for P0.19.	0
20	P0.20EF ^[1]		Enable falling edge interrupt for P0.20.	0
21	P0.21EF ^[1]		Enable falling edge interrupt for P0.21.	0
22	P0.22EF		Enable falling edge interrupt for P0.22.	0
23	P0.23EF ^[1]		Enable falling edge interrupt for P0.23.	0
24	P0.24EF ^[1]		Enable falling edge interrupt for P0.24.	0
25	P0.25EF		Enable falling edge interrupt for P0.25.	0
26	P0.26EF		Enable falling edge interrupt for P0.26.	0
27	P0.27EF ^[1]		Enable falling edge interrupt for P0.27.	0

Table 117. GPIO Interrupt Enable for port 0 Falling Edge (IO0IntEnF - address 0x4002 8094) bit description ...continued

Bit	Symbol	Value	Description	Reset value
28	P0.28EF ^[1]		Enable falling edge interrupt for P0.28.	0
29	P0.29EF		Enable falling edge interrupt for P0.29.	0
30	P0.30EF		Enable falling edge interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.5 GPIO Interrupt Enable for port 2 Falling Edge (IO2IntEnF - 0x4002 80B4)

Each bit in these read-write registers enables the falling edge interrupt for the corresponding GPIO port 2 pin.

Table 118. GPIO Interrupt Enable for port 2 Falling Edge (IO2IntEnF - 0x4002 80B4) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0EF		Enable falling edge interrupt for P2.0	0
		0	Falling edge interrupt is disabled on P2.0.	
		1	Falling edge interrupt is enabled on P2.0.	
1	P2.1EF		Enable falling edge interrupt for P2.1.	0
2	P2.2EF		Enable falling edge interrupt for P2.2.	0
3	P2.3EF		Enable falling edge interrupt for P2.3.	0
4	P2.4EF		Enable falling edge interrupt for P2.4.	0
5	P2.5EF		Enable falling edge interrupt for P2.5.	0
6	P2.6EF		Enable falling edge interrupt for P2.6.	0
7	P2.7EF		Enable falling edge interrupt for P2.7.	0
8	P2.8EF		Enable falling edge interrupt for P2.8.	0
9	P2.9EF		Enable falling edge interrupt for P2.9.	0
10	P2.10EF		Enable falling edge interrupt for P2.10.	0
11	P2.11EF ^[1]		Enable falling edge interrupt for P2.11.	0
12	P2.12EF ^[1]		Enable falling edge interrupt for P2.12.	0
13	P2.13EF ^[1]		Enable falling edge interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.6 GPIO Interrupt Status for port 0 Rising Edge Interrupt (IO0IntStatR - 0x4002 8084)

Each bit in these read-only registers indicates the rising edge interrupt status for port 0.

Table 119. GPIO Interrupt Status for port 0 Rising Edge Interrupt (IO0IntStatR - 0x4002 8084) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0REI		Status of Rising Edge Interrupt for P0.0	0
		0	A rising edge has not been detected on P0.0.	
		1	Interrupt has been generated due to a rising edge on P0.0.	
1	P0.1REI		Status of Rising Edge Interrupt for P0.1.	0
2	P0.2REI		Status of Rising Edge Interrupt for P0.2.	0
3	P0.3REI		Status of Rising Edge Interrupt for P0.3.	0
4	P0.4REI ^[1]		Status of Rising Edge Interrupt for P0.4.	0
5	P0.5REI ^[1]		Status of Rising Edge Interrupt for P0.5.	0
6	P0.6REI		Status of Rising Edge Interrupt for P0.6.	0
7	P0.7REI		Status of Rising Edge Interrupt for P0.7.	0
8	P0.8REI		Status of Rising Edge Interrupt for P0.8.	0
9	P0.9REI		Status of Rising Edge Interrupt for P0.9.	0
10	P0.10REI		Status of Rising Edge Interrupt for P0.10.	0
11	P0.11REI		Status of Rising Edge Interrupt for P0.11.	0
14:12	-		Reserved.	NA
15	P0.15REI		Status of Rising Edge Interrupt for P0.15.	0
16	P0.16REI		Status of Rising Edge Interrupt for P0.16.	0
17	P0.17REI		Status of Rising Edge Interrupt for P0.17.	0
18	P0.18REI		Status of Rising Edge Interrupt for P0.18.	0
19	P0.19REI ^[1]		Status of Rising Edge Interrupt for P0.19.	0
20	P0.20REI ^[1]		Status of Rising Edge Interrupt for P0.20.	0
21	P0.21REI ^[1]		Status of Rising Edge Interrupt for P0.21.	0
22	P0.22REI		Status of Rising Edge Interrupt for P0.22.	0
23	P0.23REI ^[1]		Status of Rising Edge Interrupt for P0.23.	0
24	P0.24REI ^[1]		Status of Rising Edge Interrupt for P0.24.	0
25	P0.25REI		Status of Rising Edge Interrupt for P0.25.	0
26	P0.26REI		Status of Rising Edge Interrupt for P0.26.	0
27	P0.27REI ^[1]		Status of Rising Edge Interrupt for P0.27.	0
28	P0.28REI ^[1]		Status of Rising Edge Interrupt for P0.28.	0
29	P0.29REI		Status of Rising Edge Interrupt for P0.29.	0
30	P0.30REI		Status of Rising Edge Interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.7 GPIO Interrupt Status for port 2 Rising Edge Interrupt (IO2IntStatR - 0x4002 80A4)

Each bit in these read-only registers indicates the rising edge interrupt status for port 2.

Table 120. GPIO Interrupt Status for port 2 Rising Edge Interrupt (IO2IntStatR - 0x4002 80A4) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0REI		Status of Rising Edge Interrupt for P2.0	0
		0	A rising edge has not been detected on P2.0.	
		1	Interrupt has been generated due to a rising edge on P2.0.	
1	P2.1REI		Status of Rising Edge Interrupt for P2.1.	0
2	P2.2REI		Status of Rising Edge Interrupt for P2.2.	0
3	P2.3REI		Status of Rising Edge Interrupt for P2.3.	0
4	P2.4REI		Status of Rising Edge Interrupt for P2.4.	0
5	P2.5REI		Status of Rising Edge Interrupt for P2.5.	0
6	P2.6REI		Status of Rising Edge Interrupt for P2.6.	0
7	P2.7REI		Status of Rising Edge Interrupt for P2.7.	0
8	P2.8REI		Status of Rising Edge Interrupt for P2.8.	0
9	P2.9REI		Status of Rising Edge Interrupt for P2.9.	0
10	P2.10REI		Status of Rising Edge Interrupt for P2.10.	0
11	P2.11REI ^[1]		Status of Rising Edge Interrupt for P2.11.	0
12	P2.12REI ^[1]		Status of Rising Edge Interrupt for P2.12.	0
13	P2.13REI ^[1]		Status of Rising Edge Interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.8 GPIO Interrupt Status for port 0 Falling Edge Interrupt (IO0IntStatF - 0x4002 8088)

Each bit in these read-only registers indicates the falling edge interrupt status for port 0.

Table 121. GPIO Interrupt Status for port 0 Falling Edge Interrupt (IO0IntStatF - 0x4002 8088) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0FEI		Status of Falling Edge Interrupt for P0.0	0
		0	A falling edge has not been detected on P0.0.	
		1	Interrupt has been generated due to a falling edge on P0.0.	
1	P0.1FEI		Status of Falling Edge Interrupt for P0.1.	0
2	P0.2FEI		Status of Falling Edge Interrupt for P0.2.	0
3	P0.3FEI		Status of Falling Edge Interrupt for P0.3.	0
4	P0.4FEI ^[1]		Status of Falling Edge Interrupt for P0.4.	0
5	P0.5FEI ^[1]		Status of Falling Edge Interrupt for P0.5.	0
6	P0.6FEI		Status of Falling Edge Interrupt for P0.6.	0
7	P0.7FEI		Status of Falling Edge Interrupt for P0.7.	0

Table 121. GPIO Interrupt Status for port 0 Falling Edge Interrupt (IO0IntStatF - 0x4002 8088)
bit description ...continued

Bit	Symbol	Value	Description	Reset value
8	P0.8FEI		Status of Falling Edge Interrupt for P0.8.	0
9	P0.9FEI		Status of Falling Edge Interrupt for P0.9.	0
10	P0.10FEI		Status of Falling Edge Interrupt for P0.10.	0
11	P0.11FEI		Status of Falling Edge Interrupt for P0.11.	0
14:12	-		Reserved.	NA
15	P0.15FEI		Status of Falling Edge Interrupt for P0.15.	0
16	P0.16FEI		Status of Falling Edge Interrupt for P0.16.	0
17	P0.17FEI		Status of Falling Edge Interrupt for P0.17.	0
18	P0.18FEI		Status of Falling Edge Interrupt for P0.18.	0
19	P0.19FEI ^[1]		Status of Falling Edge Interrupt for P0.19.	0
20	P0.20FEI ^[1]		Status of Falling Edge Interrupt for P0.20.	0
21	P0.21FEI ^[1]		Status of Falling Edge Interrupt for P0.21.	0
22	P0.22FEI		Status of Falling Edge Interrupt for P0.22.	0
23	P0.23FEI ^[1]		Status of Falling Edge Interrupt for P0.23.	0
24	P0.24FEI ^[1]		Status of Falling Edge Interrupt for P0.24.	0
25	P0.25FEI		Status of Falling Edge Interrupt for P0.25.	0
26	P0.26FEI		Status of Falling Edge Interrupt for P0.26.	0
27	P0.27FEI ^[1]		Status of Falling Edge Interrupt for P0.27.	0
28	P0.28FEI ^[1]		Status of Falling Edge Interrupt for P0.28.	0
29	P0.29FEI		Status of Falling Edge Interrupt for P0.29.	0
30	P0.30FEI		Status of Falling Edge Interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.9 GPIO Interrupt Status for port 2 Falling Edge Interrupt (IO2IntStatF - 0x4002 80A8)

Each bit in these read-only registers indicates the falling edge interrupt status for port 2.

Table 122. GPIO Interrupt Status for port 2 Falling Edge Interrupt (IO2IntStatF - 0x4002 80A8)
bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0FEI		Status of Falling Edge Interrupt for P2.0	0
		0	A falling edge has not been detected on P2.0.	
		1	Interrupt has been generated due to a falling edge on P2.0.	
1	P2.1FEI		Status of Falling Edge Interrupt for P2.1.	0
2	P2.2FEI		Status of Falling Edge Interrupt for P2.2.	0
3	P2.3FEI		Status of Falling Edge Interrupt for P2.3.	0
4	P2.4FEI		Status of Falling Edge Interrupt for P2.4.	0
5	P2.5FEI		Status of Falling Edge Interrupt for P2.5.	0
6	P2.6FEI		Status of Falling Edge Interrupt for P2.6.	0

Table 122. GPIO Interrupt Status for port 2 Falling Edge Interrupt (IO2IntStatF - 0x4002 80A8) bit description ...continued

Bit	Symbol	Value	Description	Reset value
7	P2.7FEI		Status of Falling Edge Interrupt for P2.7.	0
8	P2.8FEI		Status of Falling Edge Interrupt for P2.8.	0
9	P2.9FEI		Status of Falling Edge Interrupt for P2.9.	0
10	P2.10FEI		Status of Falling Edge Interrupt for P2.10.	0
11	P2.11FEI ^[1]		Status of Falling Edge Interrupt for P2.11.	0
12	P2.12FEI ^[1]		Status of Falling Edge Interrupt for P2.12.	0
13	P2.13FEI ^[1]		Status of Falling Edge Interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.10 GPIO Interrupt Clear register for port 0 (IO0IntClr - 0x4002 808C)

Writing a 1 into a bit in this write-only register clears any interrupts for the corresponding port 0 pin.

Table 123. GPIO Interrupt Clear register for port 0 (IO0IntClr - 0x4002 808C) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0CI		Clear GPIO port Interrupts for P0.0	0
		0	Corresponding bits in IOxIntStatR and IOxIntStatF are unchanged.	
		1	Corresponding bits in IOxIntStatR and IOxStatF are cleared.	
1	P0.1CI		Clear GPIO port Interrupts for P0.1.	0
2	P0.2CI		Clear GPIO port Interrupts for P0.2.	0
3	P0.3CI		Clear GPIO port Interrupts for P0.3.	0
4	P0.4CI ^[1]		Clear GPIO port Interrupts for P0.4.	0
5	P0.5CI ^[1]		Clear GPIO port Interrupts for P0.5.	0
6	P0.6CI		Clear GPIO port Interrupts for P0.6.	0
7	P0.7CI		Clear GPIO port Interrupts for P0.7.	0
8	P0.8CI		Clear GPIO port Interrupts for P0.8.	0
9	P0.9CI		Clear GPIO port Interrupts for P0.9.	0
10	P0.10CI		Clear GPIO port Interrupts for P0.10.	0
11	P0.11CI		Clear GPIO port Interrupts for P0.11.	0
14:12	-		Reserved.	NA
15	P0.15CI		Clear GPIO port Interrupts for P0.15.	0
16	P0.16CI		Clear GPIO port Interrupts for P0.16.	0
17	P0.17CI		Clear GPIO port Interrupts for P0.17.	0
18	P0.18CI		Clear GPIO port Interrupts for P0.18.	0
19	P0.19CI ^[1]		Clear GPIO port Interrupts for P0.19.	0
20	P0.20CI ^[1]		Clear GPIO port Interrupts for P0.20.	0
21	P0.21CI ^[1]		Clear GPIO port Interrupts for P0.21.	0

Table 123. GPIO Interrupt Clear register for port 0 (IO0IntClr - 0x4002 808C) bit description ...continued

Bit	Symbol	Value	Description	Reset value
22	P0.22CI		Clear GPIO port Interrupts for P0.22.	0
23	P0.23CI ^[1]		Clear GPIO port Interrupts for P0.23.	0
24	P0.24CI ^[1]		Clear GPIO port Interrupts for P0.24.	0
25	P0.25CI		Clear GPIO port Interrupts for P0.25.	0
26	P0.26CI		Clear GPIO port Interrupts for P0.26.	0
27	P0.27CI ^[1]		Clear GPIO port Interrupts for P0.27.	0
28	P0.28CI ^[1]		Clear GPIO port Interrupts for P0.28.	0
29	P0.29CI		Clear GPIO port Interrupts for P0.29.	0
30	P0.30CI		Clear GPIO port Interrupts for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

9.5.6.11 GPIO Interrupt Clear register for port 2 (IO2IntClr - 0x4002 80AC)

Writing a 1 into a bit in this write-only register clears any interrupts for the corresponding port 2 pin.

Table 124. GPIO Interrupt Clear register for port 2 (IO2IntClr - 0x4002 80AC) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0CI		Clear GPIO port Interrupts for P2.0	0
		0	Corresponding bits in IOxIntStatR and IOxIntStatF are unchanged.	
		1	Corresponding bits in IOxIntStatR and IOxStatF are cleared.	
1	P2.1CI		Clear GPIO port Interrupts for P2.1.	0
2	P2.2CI		Clear GPIO port Interrupts for P2.2.	0
3	P2.3CI		Clear GPIO port Interrupts for P2.3.	0
4	P2.4CI		Clear GPIO port Interrupts for P2.4.	0
5	P2.5CI		Clear GPIO port Interrupts for P2.5.	0
6	P2.6CI		Clear GPIO port Interrupts for P2.6.	0
7	P2.7CI		Clear GPIO port Interrupts for P2.7.	0
8	P2.8CI		Clear GPIO port Interrupts for P2.8.	0
9	P2.9CI		Clear GPIO port Interrupts for P2.9.	0
10	P2.10CI		Clear GPIO port Interrupts for P2.10.	0
11	P2.11CI ^[1]		Clear GPIO port Interrupts for P2.11.	0
12	P2.12CI ^[1]		Clear GPIO port Interrupts for P2.12.	0
13	P2.13CI ^[1]		Clear GPIO port Interrupts for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

9.6 GPIO usage notes

9.6.1 Example: An instantaneous output of 0s and 1s on a GPIO port

Solution 1: using 32-bit (word) accessible fast GPIO registers

```
FI00MASK = 0xFFFF00FF ;  
FI00PIN  = 0x0000A500;
```

Solution 2: using 16-bit (half-word) accessible fast GPIO registers

```
FI00MASKL = 0x00FF;  
FI00PINL  = 0xA500;
```

Solution 3: using 8-bit (byte) accessible fast GPIO registers

```
FI00PIN1  = 0xA5;
```

9.6.2 Writing to FIOSET/FIOCLR vs. FIOPIN

Writing to the FIOSET/FIOCLR registers allow a program to easily change a port's output pin(s) to both high and low levels at the same time. When FIOSET or FIOCLR are used, only pin/bit(s) written with 1 will be changed, while those written as 0 will remain unaffected.

Writing to the FIOPIN register enables instantaneous output of a desired value on the parallel GPIO. Data written to the FIOPIN register will affect all pins configured as outputs on that port: zeroes in the value will produce low level pin outputs and ones in the value will produce high level pin outputs.

A subset of a port's pins may be changed by using the FIOMASK register to define which pins are affected. FIOMASK is set up to contain zeroes in bits corresponding to pins that will be changed, and ones for all others. Solution 2 from [Section 9.6.1](#) above illustrates output of 0xA5 on PORT0 pins 15 to 8 while preserving all other PORT0 output pins as they were before.